

Notes: Crawford & Sobel (Econometrica, 1982)

Strategic Information Transmission

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1 Big picture

- **Question.** How much information can be transmitted in equilibrium when communication is costless, unverifiable, and strategically biased?
- **Setting.** A better-informed Sender observes a scalar state/type m and sends a message to a Receiver. The Receiver chooses an action y that affects both players' payoffs.
- **Key friction.** The Sender and Receiver both care about the Receiver's action, but their ideal actions differ. The parameter b measures the conflict of interest.
- **Central result.** Equilibrium communication takes a *partition* form: the Sender does not report the exact state, but reveals which interval of the state space contains the state.
- **Information content.** Messages are informative but coarse. The more aligned the Sender and Receiver are, the finer the equilibrium partition can be.
- **Multiple equilibria.** There is usually a completely uninformative babbling equilibrium and also more informative partition equilibria, up to a maximum number of steps $N(b)$.
- **Equilibrium selection.** The paper argues that the finest partition equilibrium is salient and attractive because, ex ante, both players prefer more informative equilibria.
- **Strategic communication insight.** Even without lying costs or exogenous signaling costs, strategic incentives limit communication. The Sender can credibly transmit only coarse information because exact revelation would give him incentives to manipulate the Receiver's action.
- **Economic takeaway.** Cheap talk is neither fully useless nor fully revealing. It transmits information precisely to the extent permitted by preference alignment.

2 Incremental contribution of the paper

- **From signaling costs to endogenous information loss.** Earlier signaling models often relied on exogenous signaling costs or sorting costs. Crawford and Sobel show that communication can be informative even when signals are costless, because the Receiver's rational expectations create endogenous limits on what messages can mean.
- **Canonical cheap-talk equilibrium concept.** The paper formalizes strategic communication as a Bayesian Nash equilibrium in message and action rules. This became the standard cheap-talk framework.
- **Partition-equilibrium theorem.** The paper proves that, under standard single-crossing-type assumptions, all equilibria are essentially equivalent to finite interval partitions of the type space.
- **Preference-alignment comparative statics.** The paper links the fineness of communication directly to the divergence of preferences: smaller bias b supports more partition elements and hence more informative communication.
- **Equilibrium-selection argument.** The paper goes beyond existence by showing that, ex ante, both Receiver and Sender prefer more informative partitions under conditions satisfied by the main example.
- **Bridge to bargaining and organizations.** The model explains why communication among self-interested agents can improve outcomes while still being systematically distorted, making it foundational for bargaining, political economy, organizational communication, lobbying, and delegation.

3 Model objects and measurement

3.1 Players, type, and messages

- There are two players: Sender S and Receiver R .

- The Sender privately observes $m \in [0, 1]$, drawn from distribution F with density f .
- The Sender sends a signal/message n to the Receiver. The signaling rule may be stochastic:

$$q(n | m).$$

- The Receiver observes the message and chooses an action $y(n)$.
- The Receiver does not observe the Sender's true type directly and does not commit ex ante to an action rule before the Sender communicates.

Interpretation: The only source of private information is the Sender's knowledge of m . The strategic problem is not whether messages are costly; it is whether the Sender's incentives make fine messages credible.

3.2 Preferences

The Sender's utility is

$$U^S(y, m, b),$$

where b measures how far the Sender's ideal action is from the Receiver's. The Receiver's utility is

$$U^R(y, m).$$

For each m , define the ideal actions

$$y^S(m, b) \equiv \arg \max_y U^S(y, m, b), \quad y^R(m) \equiv \arg \max_y U^R(y, m).$$

- Both utility functions are concave in y .
- The best action from each player's viewpoint is increasing in m .
- In the main quadratic example, the Sender's ideal action is shifted upward by b relative to the Receiver's.

Interpretation: The bias parameter b is the key comparative-static variable. If $b = 0$, preferences coincide and full revelation is possible. If b is large, only coarse or no information can be credibly transmitted.

3.3 Bayesian Nash equilibrium in communication rules

An equilibrium consists of a signaling rule $q(n | m)$ and an action rule $y(n)$ such that:

1. each Sender type chooses a message distribution that maximizes the Sender's expected utility, given the Receiver's action rule;
2. the Receiver chooses the optimal action after each message, using Bayes' rule to form posterior beliefs about m .

Formally, the Receiver's posterior after message n is proportional to

$$q(n | m)f(m).$$

The Receiver's action solves

$$y(n) \in \arg \max_y \int_0^1 U^R(y, m)p(m | n) dm.$$

Interpretation: The Receiver is fully rational. The model does not assume that the Receiver believes messages literally. The Receiver interprets messages through the equilibrium signaling rule.

4 Models and theoretical specifications

4.1 Why exact truth-telling generally fails

If the Sender's ideal action differs from the Receiver's, a literal report of m would induce R to choose $y^R(m)$, while S would prefer $y^S(m, b)$. The Sender would therefore want to misreport toward types that induce a more favorable action. Hence, exact truth-telling is not credible unless preferences coincide.

Interpretation: Cheap talk differs from verifiable disclosure. The message has no direct payoff consequence; it matters only through the action it induces. Therefore, the Sender wants the Receiver to infer just enough information to move the action in a useful direction, but not enough to pin down the true state.

4.2 Finite action support: Lemma 1

Lemma 1 says that if $y^S(m, b) \neq y^R(m)$ for all m , then actions induced in equilibrium are separated by a positive distance and the set of actions induced in equilibrium is finite.

Interpretation: This is the first key step. If two induced actions were arbitrarily close, some Sender type near the boundary would be able to profitably switch messages. The single-crossing logic prevents a continuum of distinct actions when preferences are biased.

4.3 Partition equilibria

A partition with N steps is

$$0 = a_0 < a_1 < \dots < a_N = 1.$$

For each interval (a_i, a_{i+1}) , the Receiver chooses the optimal action conditional on m lying in that interval:

$$\bar{y}(a_i, a_{i+1}) \equiv \arg \max_y \int_{a_i}^{a_{i+1}} U^R(y, m) f(m) dm.$$

The marginal Sender type at boundary a_i must be indifferent between inducing the Receiver's action in the lower interval and the upper interval:

$$U^S(\bar{y}(a_i, a_{i+1}), a_i, b) - U^S(\bar{y}(a_{i-1}, a_i), a_i, b) = 0.$$

Interpretation: Each boundary is pinned down by Sender indifference. Within each cell, all types pool on the same effective message. Communication is therefore coarse but credible.

4.4 Theorem 1: all equilibria are partition equilibria

Theorem 1 establishes three results:

- for every N between 1 and a maximum $N(b)$, at least one N -step partition equilibrium exists;
- $N = 1$ is the completely uninformative babbling equilibrium;
- every equilibrium is essentially equivalent to one of these finite partition equilibria.

Interpretation: This is the core theorem. The model does not restrict the Sender's messages to be intervals, but equilibrium forces that structure. Arbitrary messages reduce to a finite number of induced actions and therefore to a finite partition of the state space.

4.5 Quadratic-uniform example

In the paper's main example, m is uniformly distributed on $[0, 1]$ and utilities are quadratic-loss utilities with a bias b . The partition equation becomes a second-order difference equation:

$$a_{i+1} = 2a_i - a_{i-1} + 4b.$$

A solution has the form

$$a_i = a_1 i + 2i(i-1)b.$$

The maximum number of equilibrium partition steps satisfies approximately

$$N(b) = \left\lceil -\frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{2}{b}} \right\rceil.$$

Interpretation: As $b \downarrow 0$, preferences approach alignment and $N(b)$ rises. As b rises, partitions become coarser; when b is large enough, only babbling remains.

4.6 Equilibrium informativeness and ex ante payoffs

For a fixed partition, more steps reduce the residual variance of m after observing the message. The Receiver benefits because conditional expectations become more precise. The Sender also benefits ex ante because, before learning his type, a more precise but still equilibrium-compatible message allows the Receiver to choose actions closer to the true state while preserving incentive compatibility.

Interpretation: After the Sender observes his type, different Sender types prefer different equilibria. But before observing the type, both players prefer finer partitions under the paper's conditions. This gives a strong selection rationale for the finest partition.

4.7 Comparative statics in preference alignment

The paper generalizes the example using monotonicity condition (M). The key comparative statics are:

- $N(b)$ is nonincreasing in b .
- For a given b , the Receiver strictly prefers partitions with more steps.
- For a given number of steps, the Receiver prefers the partition associated with more similar preferences.
- The Sender also strictly prefers more steps ex ante.

Interpretation: Greater preference similarity means the Sender can credibly reveal more. The Receiver's equilibrium expected utility rises not because the Receiver trusts messages more naively, but because the equilibrium messages contain more information.

5 Theoretical design and credibility of the mechanism

5.1 What makes the partition result work

- **One-dimensional private information.** The state m is scalar and ordered.
- **Single-crossing structure.** The Sender's ideal action moves monotonically with m and differs systematically from the Receiver's.

- **Receiver rational expectations.** The Receiver interprets messages using the equilibrium distribution of messages conditional on types.
- **No commitment.** The Receiver cannot commit to non-optimal responses to make truth telling incentive compatible.
- **Costless messages.** Since messages are cheap talk, all credibility comes from equilibrium incentives rather than physical or monetary costs.

Interpretation: These assumptions isolate the strategic-communication mechanism. If messages were verifiable or if the Receiver could commit, the partition logic would change.

5.2 Why preference similarity matters

When b is small, a boundary type is nearly indifferent between neighboring induced actions. This allows more intervals to be sustained without creating profitable deviations. When b is large, neighboring actions must be far apart to preserve incentive compatibility, so only coarse messages can survive.

Interpretation: Preference alignment operates like a credibility technology. It makes fine communication self-enforcing.

5.3 What the paper cannot fully prove

- The paper focuses on scalar private information. Multidimensional types are much harder and do not generally lead to simple interval partitions.
- It does not model repeated communication, reputation, reputational costs, or lying costs.
- Multiple equilibria remain. The finest partition is motivated by salience and ex ante welfare, not uniquely selected by the static game alone.
- The model abstracts from commitment devices, delegation mechanisms, and contracts that could change communication incentives.

Interpretation: The paper is foundational because it gives a tractable benchmark. Later cheap-talk, persuasion, delegation, and information-design models often start by modifying one of these assumptions.

6 Theorem blocks and figure from the paper

6.1 Equilibrium conditions and Lemma 1: Bayesian Nash communication and finite induced actions

Read:

- The equilibrium conditions define Sender optimality and Receiver Bayesian optimality.
- The Sender chooses messages to induce preferred Receiver actions; the Receiver updates beliefs using the message rule and chooses an optimal action.
- Lemma 1 establishes that if Sender and Receiver ideal actions differ everywhere, equilibrium cannot induce a continuum of actions.
- Therefore, any informative equilibrium must collapse types into a finite set of message-induced actions.

Economic message: Lemma 1 is the technical bridge from arbitrary cheap-talk messages to finite partitions. It tells us that communication must be noisy/coarse whenever preferences are not perfectly aligned.

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$q(n|m)$, and an action rule for R , denoted $y(n)$, such that:

- (1) for each $m \in [0, 1]$, $\int_N q(n|m) dn = 1$, where the Borel set N is the set of feasible signals, and if n^* is in the support of $q(\cdot|m)$, then n^* solves $\max_{n \in N} U^S(y(n), m, b)$; and
- (2) for each n , $y(n)$ solves $\max_y \int_0^1 U^R(y, m)p(m|n) dm$,
where $p(m|n) \equiv q(n|m)f(m) / \int_0^1 q(n|t)f(t) dt$.²

Condition (1) says that S 's signaling rule yields an expected-utility maximizing action for each of his information "types," taking R 's action rule as given. Condition (2) says that R responds optimally to each possible signal, using Bayes' Rule to update his prior, taking into account S 's signaling strategy and the signal

(a) Equilibrium conditions: Sender rule and Receiver best response

equilibrium. Define, for all $m \in [0, 1]$,

$$(3) \quad y^S(m, b) \equiv \arg \max U^S(y, m, b)$$

and

$$(4) \quad y^R(m) \equiv \arg \max U^R(y, m),$$

where $\arg \max U^S(y, m, b)$, for example, denotes the value of y that maximizes $U^S(y, m, b)$. Since $U_{11}^i(\cdot) < 0$ and $U_{12}^i(\cdot) > 0$, $i = R, S$, $y^S(m, b)$ and $y^R(m)$ are well defined and continuous in m .

LEMMA 1: *If $y^S(m, b) \neq y^R(m)$ for all m , then there exists an $\epsilon > 0$ such that if u and v are actions induced in equilibrium, $|u - v| \geq \epsilon$. Further, the set of actions induced in equilibrium is finite.*

PROOF: Let u and v , with $u < v$, be two actions induced in equilibrium. Since an S -type who induces u (v) thereby reveals a weak preference for that action over v (u), by continuity there exists an $\bar{m} \in [0, 1]$ such that $U^S(u, \bar{m}, b) = U^S(v, \bar{m}, b)$. Since $U_{11}^S(\cdot) < 0$ and $U_{12}^S(\cdot) > 0$, it follows from this that

$$(5) \quad u < y^S(\bar{m}, b) < v,$$

$$(6) \quad u \text{ is not induced by any } S\text{-type } m > \bar{m}, \text{ and}$$

$$(7) \quad v \text{ is not induced by any } S\text{-type } m < \bar{m}.$$

In turn, (6), (7), and our assumption that $U_{12}^R(\cdot) > 0$ imply that

$$(8) \quad u \leq y^R(\bar{m}) \leq v.$$

However, if $y^R(m) \neq y^S(m, b)$ for all $m \in [0, 1]$, there is an $\epsilon > 0$ such that $|y^R(m) - y^S(m, b)| \geq \epsilon$ for all $m \in [0, 1]$. It follows from (5) and (8) that $v - u \geq \epsilon$. Since the set of actions induced in equilibrium is bounded by $y^R(0)$ and $y^R(1)$ because $U_{12}^R(\cdot) > 0$, this completes the proof. Q.E.D.

(b) Lemma 1: finite set of induced actions

6.2 Theorem 1 and quadratic example: partition-equilibrium characterization

Read:

- Theorem 1 states that there are partition equilibria of sizes 1 through $N(b)$.
- The boundary conditions make the boundary type indifferent between adjacent induced actions.
- The theorem also says that every equilibrium is essentially equivalent to one of these partition equilibria.
- In the quadratic-uniform example, the partition recursion is explicit and $N(b)$ can be calculated from the bias parameter.

Economic message: Theorem 1 is the paper's central contribution: cheap talk endogenously becomes interval disclosure. The quadratic example makes the comparative statics transparent: smaller conflict supports finer communication.

Now we are ready to state Theorem 1, which establishes the existence of equilibria, and characterizes them.

THEOREM 1: Suppose b is such that $y^S(m, b) \neq y^R(m)$ for all m . Then there exists a positive integer $N(b)$ such that, for every N with $1 \leq N \leq N(b)$, there exists at least one equilibrium $(y(n), q(n|m))$, where $q(n|m)$ is uniform, supported on $[a_i, a_{i+1}]$ if $m \in (a_i, a_{i+1})$,

$$(A) \quad U^S(\bar{y}(a_i, a_{i+1}), a_i, b) - U^S(\bar{y}(a_{i-1}, a_i), a_i, b) = 0 \quad (i = 1, \dots, N-1),$$

$$(10) \quad y(n) = \bar{y}(a_i, a_{i+1}) \quad \text{for all } n \in (a_i, a_{i+1}),$$

$$(11) \quad a_0 = 0, \quad \text{and}$$

$$(12) \quad a_N = 1.$$

Further, any equilibrium is essentially³ equivalent to one in this class, for some value of N with $1 \leq N \leq N(b)$.

(a) Theorem 1: existence and characterization of partition equilibria

The arbitrage condition (A) specializes to

$$(20) \quad -\left(\frac{a_i + a_{i+1}}{2} - a_i - b\right)^2 = -\left(\frac{a_{i-1} + a_i}{2} - a_i - b\right)^2 \quad (i = 1, \dots, N-1)$$

which can only hold, given the monotonicity of a , if

$$(21) \quad a_{i+1} = 2a_i - a_{i-1} + 4b \quad (i = 1, \dots, N-1)$$

This second-order linear difference equation has a class of solutions parametrized by a_1 (given that $a_0 = 0$):

$$(22) \quad a_i = a_1 i + 2i(i-1)b \quad (i = 1, \dots, N)$$

$N(b)$ in Theorem 1 is the largest positive integer i such that $2i(i-1)b < 1$, which is easily shown to be

$$\left\langle -\frac{1}{2} + \frac{1}{2} \left(1 + \frac{2}{b}\right)^{1/2} \right\rangle$$

(where $\langle z \rangle$ denotes the smallest integer greater than or equal to z). Thus, it is

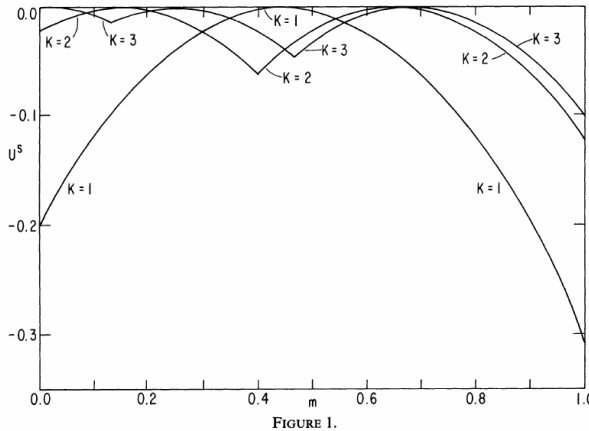
(b) Quadratic-uniform example: partition recursion and $N(b)$

6.3 Figure 1 and Theorem 2: Sender type payoffs and monotone comparative statics

Read:

- Figure 1 plots Sender payoffs across types for partitions with $K = 1, 2, 3$ in the example with $b = 1/20$.
- Different Sender types may prefer different equilibria after learning their type.
- Ex ante, before the Sender learns his type, the finer partition is more attractive.
- Theorem 2 gives sufficient conditions under which the monotonicity condition used for comparative statics holds beyond the quadratic example.

Economic message: Figure 1 explains why ex post bargaining over equilibrium selection is difficult: types have conflicting preferences over which equilibrium should be played. Theorem 2 helps justify focusing on the finest equilibrium ex ante.



(a) Figure 1: Sender payoffs under different partition sizes

(b) Theorem 2: sufficient condition for monotone comparative statics

THEOREM 2: For a given value of b , if $U_2^S(y, a, b) + U_1^S(y, a, b)$ is nondecreasing in y and $\int_0^a U_{11}^R(y, m) f(m) dm + U_1^R(y, a) f(a)$ is nonincreasing in a , then all solutions to (A) satisfy condition (M).

PROOF: Let $a \in [0, 1]$ be given. To study how solutions to (A) change when

6.4 Lemma 6 and Theorems 3–5: preference similarity, more steps, and ex ante selection

Read:

- Lemma 6 says the maximum possible equilibrium partition size $N(b)$ is nonincreasing in the degree of preference conflict b .
- Theorem 3 says the Receiver strictly prefers equilibria with more steps for fixed preferences.
- Theorem 4 says, for a fixed number of steps, the Receiver prefers the partition associated with more similar preferences.
- Theorem 5 says the Sender also strictly prefers more steps ex ante, before learning his type.

Economic message: The welfare case for the finest partition is strong before private information is realized. More communication is jointly valuable ex ante, even though strategic bias prevents full revelation.

Therefore the proof is omitted.

LEMMA 6: *The maximum possible equilibrium partition size, $N(b)$, is nonincreasing in b , the difference between agents' preferences.*

PROOF: Suppose $b' < b$. Let $a(N(b), b)$ be a partition equilibrium of size $N(b)$, and let $\bar{a}(N(b), b')$ be the partial partition satisfying (A) with $\bar{a}_i(N(b), b') = a_i(N(b), b)$. By Lemma 5, $\bar{a}_i(N(b), b') < a_i(N(b), b)$ for all $i = 2, \dots, N(b)$. In particular, $\bar{a}(N(b), b')$ is at least of length $N(b)$. It follows that $N(b') \geq N(b)$. *Q.E.D.*

We are now ready to generalize the comparative statics results of Section 4.

THEOREM 3: *For given preferences (i.e., b), R always strictly prefers equilibrium partitions with more steps (larger N 's).*

(a) Lemma 6 and Theorem 3: maximum partition size and Receiver preference for more steps

THEOREM.

Q.E.D.

THEOREM 4: *For a given number of steps (i.e., N), R always prefers the equilibrium partition associated with more similar preferences (i.e., a smaller value of b).*

The proof of Theorem 4 is a straightforward application of Lemma 5 and an argument like that used to prove Theorem 3, and is therefore omitted.

THEOREM 5: *For given preferences (i.e., b), S always strictly prefers ex ante (that is, before learning his type) equilibrium partitions with more steps (larger N 's).*

PROOF: Maintaining the notation used in the proof of Theorem 3.

(b) Theorems 4–5: preference similarity and Sender ex ante preference

7 Short summary

- Crawford and Sobel create the canonical cheap-talk model of strategic communication.
- A Sender observes a scalar state and sends a costless message to a Receiver, who then chooses an action.
- The Sender and Receiver have related but biased preferences over the action.
- Exact revelation is generally not credible because the Sender wants to push the Receiver's action toward his own ideal point.
- Equilibrium communication takes a partition form: messages reveal only intervals of the state space.
- There is always a babbling equilibrium, and there may also be informative partition equilibria.
- The maximum number of partition steps is lower when the conflict of interest is larger.
- In the quadratic example, the partition boundaries solve a simple difference equation and $N(b)$ can be computed explicitly.
- The finest partition is attractive because both players prefer more informative equilibria ex ante under the paper's conditions.
- The paper's main message is that strategic cheap talk can transmit information, but only coarsely and only when incentives are sufficiently aligned.

8 Conclusion

8.1 Core conclusion

Crawford and Sobel show that costless communication between self-interested agents can be informative, but it cannot generally be fully revealing. In equilibrium, the Sender communicates through a finite partition of the state space. The Receiver learns which interval contains the Sender’s information and responds optimally to the conditional distribution within that interval.

8.2 Incremental contribution

The paper’s incremental contribution is to turn strategic communication into a rigorous equilibrium object. It does not rely on exogenous signaling costs, verifiability, or commitment. Instead, it shows that the credibility of messages is determined entirely by the Sender’s incentive compatibility and the Receiver’s Bayesian interpretation of messages. This created the benchmark framework for cheap talk and made it possible to ask when communication is informative, how informative it is, and how preference alignment changes information transmission.

8.3 Economic intuition

The Sender wants the Receiver to take an action that depends on the true state, but he also wants to bias that action in his preferred direction. The Receiver understands this incentive. Therefore, a message can carry information only when all Sender types assigned to that message prefer the same induced response to neighboring responses. This logic creates interval pooling. A boundary type must be indifferent between the two adjacent induced actions, which pins down the partition.

8.4 Why the finest partition matters

The finest partition is not uniquely forced by the static game, because coarser partitions are also equilibria. But it is economically salient. Before the Sender knows his type, both Sender and Receiver benefit from lower residual uncertainty. This gives a coordination argument for the equilibrium with the greatest number of cells. The paper therefore provides not only a characterization of equilibria, but also a principled reason to focus on the most informative one.

8.5 Broader implications

The results apply naturally to bargaining, lobbying, managerial communication, expert advice, and political recommendations. Whenever one party has information and another controls the decision, communication will be more effective when preferences are aligned and less effective when the informed party has strong incentives to distort. This insight remains central in modern information economics.

8.6 Limitations and extensions

The scalar-state assumption is central. In multidimensional environments, partitions need not be ordered intervals and equilibrium communication can be more complex. The model also abstracts from repeated interactions, reputational concerns, verifiable evidence, costly lying, institutional commitment, and delegation. Each of these features can expand or alter the scope for communication. The value of the Crawford–Sobel benchmark is that it identifies the pure incentive-compatibility limit of cheap talk before those additional mechanisms are added.

8.7 Takeaway

The enduring takeaway is: communication is possible without commitment, but not unconstrained. The amount of information transmitted is an equilibrium outcome governed by preference conflict. Cheap talk is informative only to the degree that the Sender's incentives make it credible.